The new INB/ELIXIR-ES portfolio of resources

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The **Spanish National Bioinformatics Institute (INB)**, the **Spanish node of ELIXIR**, renewed its Service Delivery Plan (SDP) in the second half of 2023. This periodical assessment allows reviewing the current offer by the node and offer an up-to-date portfolio of resources to ELIXIR, directly impacting the Life Sciences community.

The renewed INB/ELIXIR-ES SDP is composed of **40 resources** offered, developed and maintained by **24 groups** in **12 institutions** across Spain.

Significant resources stand out as cross-node or multi-node services, and some are part of the ELIXIR Key Service Collections as ELIXIR Core Data Resources and ELIXIR Recommended Interoperability Resources. Interestingly, this is the first time that data collections, e.g. Collaborative Spanish Variant Server (CSVS) and Genomes for Life - Cohort Study of the Genomes of Catalonia (GCAT), have been included as part of the INB/ELIXIR-ES portfolio highlighting the value and possibilities of such resources for the whole research community.

16 tools

- **Beyondcell** – *CNIO*
- **DREIMT** – *CNIO*
- **Nextflow** – *CRG*
- **rbbt** Ruby bioinformatics toolkit – *BSC*
- **BioBB** BioExcel Building Blocks – *IRB Barcelona, BSC*
- **PanDrugs** – *CNIO*
- **SQANTI** – *I2SysBio (CSIC)*
- **NLP4BIA tools** BSC Text Mining Unit tools – *BSC*
- **MBROLE** Metabolites Biological Role – *CNB (CSIC)*
- **ETE Toolkit** – *CBGP (CSIC)*
- **IntOGen** – *IRB Barcelona*
- **Redundans** – *BSC, IRB Barcelona*
- **T-Coffee** – *CRG*
- **GET_HOMOLOGUES** – *EEAD (CSIC)*
- **RSAT Plants** – *EEAD (CSIC)*
- **WfExS-backend** Workflow Execution Service – *BSC*

4 platforms

- **FlexPortal** INB Integrated platform for macromolecular flexibility – *IRB Barcelona, BSC*
- **PMut** Annotations of Pathological Mutations on Proteins – *BSC, IRB Barcelona*
- **OpenEBench** – *BSC*
- **FAIR Cookbook** – *SAS, BSC*

ELIXIR Recommended Interoperability Resources

6 databases

- **eggNOG** – *CBGP (CSIC)*
- **footprintDB** – *EEAD (CSIC)*
- **EGA** European Genome-phenome Archive – *CRG* (ELIXIR Core Data Resource)
- **CANTEMIST** – *BSC*
- **MetaPhors** – *BSC, IRB Barcelona*
- **PhylomeDB** – *BSC, IRB Barcelona*

9 web-services

- **3Did** 3D interacting domains – *IRB Barcelona*
- **APID Interactomes** – *IBMCC (CSIC)*
- **EGA Beacon** – *CRG*
- **dSysMap** Mapping human disease-mutations at systemic level – *IRB Barcelona*
- **APPRIS** – *CNIO*
- **Hipathia suite** – *FPS*
- **FAIRTracks Validation Service** – *BSC*
- **3D BIONOTES-WS and COVID19 Structural Hub** – *CNB (CSIC)*
- **Interactome3D** – *IRB Barcelona*

3 workflows

- **Master Of Pores** – *CRG*
- **Sqtlseeker2-nf** – *CRG*
- **manta/mvgwas-nf** – *CRG*

2 data collections

- **CSVS** Collaborative Spanish Variant Server – *FPS*
- **GCAT** Genomes for Life – *IGTP*

Selection process for the portfolio renovation

Expressions of Interest in joining the INB/ELIXIR-ES resources portfolio were evaluated by an independent Scientific Advisory Board (SAB) panel.

Definition of Selection Criteria → **Appointment of SAB** → **Open call for Expressions of Interest** → **SAB evaluation process** → **Analysis of the SAB review & discussion of the results** → **Rebuttal process** → **Final selection of INB/ELIXIR-ES portfolio**

- ★ Alignment with ELIXIR
- ★ Uptake of the Open Science principles
- ★ Technical excellence & scientific impact
- ★ Sustainability
- ★ Community uptake

The selection process allowed the inclusion of new resources and the renewal of those already present in the previous SDP.

Gender balance was considered when configuring the portfolio of the node.

After the SAB advised on which services to be considered in the portfolio, the resources went through an acceptance confirmation process conducted by the Institute of Health Carlos III (ISCIII).

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